

Recent Advances and Remaining Gaps in Artificial Intelligence for Cancer Drug Repurposing

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Abstract—Artificial intelligence (AI) is becoming an important tool in cancer drug repurposing. Drug repurposing means finding new cancer uses for already approved or previously studied drugs. This approach can reduce time, cost, and risk compared with traditional drug discovery. However, many AI-predicted drug candidates are still not clinically reliable because they are often based on limited datasets, incomplete drug-target information, and weak biological validation. This paper summarizes recent AI-based methods in cancer drug repurposing, compares twenty related studies, identifies major technical gaps, and proposes a hybrid validation pipeline for future research.

Index Terms—Artificial intelligence, cancer, drug repurposing, machine learning, deep learning, precision oncology, drug sensitivity.

I. INTRODUCTION

Cancer drug discovery is slow, expensive, and risky. Developing a completely new drug can take many years, and many drug candidates fail before reaching patients. Because of this, drug repurposing has become an important research direction. Drug repurposing means finding new cancer uses for drugs that already exist. These drugs may already have known safety, dosage, and toxicity information, so they may move faster into clinical testing than brand-new drugs. Recent studies show that artificial intelligence can help this process by searching large datasets, finding hidden drug–disease links, predicting drug sensitivity, and suggesting possible new uses for approved drugs [1]–[4].

However, cancer is not one simple disease. Tumors differ between cancer types, between patients, and even between cells inside the same tumor. This makes AI-based drug repurposing difficult. A model may predict that a drug will work, but the drug may fail because the model was trained on limited cell-line data, incomplete drug-target information, or bulk tumor data that hides rare resistant cell groups. Newer methods such as DrugReSC and ResisenseNet try to solve this by using single-cell data, gene-expression data, molecular descriptors, transcription-factor information, and deep learning [12], [20]. Still, the main research problem remains: AI can predict many possible repurposed cancer drugs, but stronger validation is needed before these predictions can be trusted clinically.

II. PROBLEM DEFINITION

The main technical bottleneck is the prediction-to-clinical-validation gap. In simple terms, many AI models are good at producing lists of possible drugs, but not all of these drugs will actually work in real patients. This happens because cancer data are complex, noisy, and incomplete. AI models may learn patterns from cell lines, public databases, or old studies, but real patient tumors are much more complicated.

Clinical impact: Solving this problem matters because cancer patients need treatments that are safe, effective, and matched to their tumor biology. If AI predictions are not reliable, researchers may waste time and money testing weak drug candidates. More importantly, patients with resistant or aggressive cancers may not benefit. A better AI-based repurposing system could help identify useful drugs faster, especially for patients who do not respond to standard therapies.

III. STATE-OF-THE-ART ANALYSIS

We can group into the several categories of the current state-of-the-art-methods. QSAR and support vector machine (SVM) models use chemical features to predict whether a drug may work against a cancer cell line. Network-based methods connect cancer genes, viral proteins, drug targets, and protein–protein interactions to identify repurposable drugs and combinations. For example, one HPV-associated cervical cancer study used network medicine and drug-target interaction analysis to identify possible drugs and combinations [11]. Transcriptomic signature-reversal methods compare cancer gene-expression patterns with drug-induced expression patterns and look for drugs that reverse cancer or resistance signatures. In breast cancer, Yu et al. used I-SPY2 data and CMap to find drugs that may reverse treatment resistance [10].

The most recent trend is to combine multiple AI methods rather than depend on one model. Single-cell methods such as DrugReSC study specific cancer cell subpopulations instead of averaging the whole tumor [12]. Hybrid computational pipelines combine machine learning, docking, molecular dynamics, and experimental validation, as shown in the tubulin inhibitor study by Neelima et al. [14]. Newer models also move toward personalization and explainability. CHANCE predicts patient-specific anticancer potential of non-oncology drugs [15]. TxGNN uses a graph foundation model for broad drug repurposing and provides interpretable reasoning paths

[17]. ResisenseNet uses a hybrid neural network to predict breast cancer drug sensitivity and support drug repurposing [20]. These studies show progress, but the field still needs stronger patient-level validation.

IV. COMPARISON TABLE

The following table summarizes twenty recent studies on

AI-based cancer drug repurposing published between 2021 and 2026. The papers are compared based on their primary methods, key contributions, and remaining research gaps. This comparison highlights current research trends, identifies common limitations, and provides the foundation for discussing future directions in AI-assisted cancer drug repurposing.

TABLE I
COMPARISON OF 20 PAPERS RELATED TO AI-BASED CANCER DRUG REPURPOSING

No.	Paper	Year	Main Method	Main Contribution	Main Gap
1	Capistrano et al. [1]	2022	Literature and clinical trial search	Found candidate drugs for cervical cancer repurposing.	Not AI-based; useful mainly as background.
2	Tanoli et al. [2]	2021	ML review and drug-target profiles	Explains AI/ML use in cancer drug repurposing.	Older than 2023–2026; use mainly as background.
3	Malla et al. [3]	2024	Review of computational and experimental methods	Shows repurposing value for resistant and metastatic cancers.	Broad review, not only AI-focused.
4	Xia et al. [4]	2024	Review of mechanisms, organoids, and combinations	Connects repurposing with tumor microenvironment and validation.	Very broad; not only computational.
5	Herráiz-Gil et al. [5]	2025	ML, DL, and AI workflow review	Summarizes AI tools for cancer drug repurposing.	Notes data quality and interpretability issues.
6	Garg et al. [6]	2024	AI-CADD, docking, QSAR, ADMET	Shows how AI improves anticancer drug discovery.	More about drug development than repurposing.
7	Cao et al. [7]	2024/2025	Structure-based, network-based, and ML methods	Gives a clear map of computational repositioning methods.	Review only; no new model tested.
8	Raju et al. [8]	2025	Broad AI oncology review	Links AI with diagnosis, therapy, and drug development.	Repurposing is only one part.
9	Dara and Ahmed [9]	2026	AI and multi-omics	Connects genomics, proteomics, metabolomics, and repurposing.	Data heterogeneity and bias remain issues.
10	Yu et al. [10]	2023	I-SPY2 and CMap signature reversal	Found fulvestrant as a possible resistance-reversing drug.	Limited laboratory validation.
11	Ahmed et al. [11]	2023	Network medicine and DTI	Found possible drugs and combinations for HPV cervical cancer.	Mostly computational validation.
12	Liu et al. [12]	2024	Single-cell RNA-seq and random forest	Targets disease-critical cell subpopulations.	Needs more clinical validation.
13	Sengupta et al. [13]	2024	QSAR and SVM	Built colorectal cancer models and suggested FDA-approved candidates.	In silico predictions need laboratory testing.
14	Neelima et al. [14]	2026	AutoQSAR, docking, MD, and experiments	Found possible tubulin binders such as omeprazole and podofilox.	Focused only on tubulin.
15	Dong et al. [15]	2025	Supervised ML and genome-derived pathways	Predicts patient-specific anticancer potential of non-oncology drugs.	Needs larger clinical validation.
16	Lee et al. [16]	2024	Ex vivo screening and interpretable ML	Finds repurposable neuroactive drugs for glioblastoma.	Cancer-type specific.
17	Huang et al. [17]	2024	Graph foundation model	Performs broad zero-shot drug repurposing.	Not cancer-specific, but method is relevant.
18	Picard et al. [18]	2024	Multi-layer networks and ML	Predicts new therapeutic targets for prostate cancer.	Predicts targets more than direct drug candidates.
19	Lenhof et al. [19]	2024	Conformal prediction and drug prioritization	Adds confidence/reliability to anticancer drug sensitivity prediction.	Supports repurposing indirectly.
20	Karampuri et al. [20]	2024	1D-CNN, LSTM, and DNN	Predicts breast cancer drug sensitivity and identifies possible repurposed drugs.	Needs future molecular dynamics and in vitro/in vivo validation.

V. TECHNICAL GAPS

The main gap across these studies is the prediction-to-clinical-validation gap. Many AI models can rank candidate drugs, but fewer studies prove that these drugs work in patient-like systems. Some studies use only public databases, cell lines, or computational scores. Other studies are stronger but focus on only one cancer type, one target, or one dataset. In simple words, AI is good at suggesting drugs, but the field still needs stronger proof that those drugs will work safely in real patients.

There are also other technical gaps. First, many models

are difficult to interpret, especially deep learning models. Clinicians need to know why a drug is recommended. Second, many datasets are incomplete or biased toward well-studied cancer types and drugs. Third, tumor heterogeneity is often ignored when models use bulk data. Fourth, many models do not include uncertainty estimation, so they do not clearly show how confident the prediction is. Finally, many studies lack testing in patient-derived organoids, resistant cell lines, animal models, or clinical trials.

VI. PROPOSED DIRECTION

A strong future direction is a hybrid AI validation pipeline. First, network medicine can be used to find biologically connected drug candidates. Second, transcriptomic signature reversal can check whether the drug can reverse cancer or resistance patterns. Third, single-cell analysis can test whether the drug targets dangerous cancer cell subpopulations. Fourth, ADMET and toxicity prediction can remove unsafe candidates. Fifth, uncertainty estimation can show which predictions are reliable. Finally, the best drugs should be tested in patient-derived organoids, resistant cancer cell lines, and eventually clinical trials.

This approach would be stronger than using only one AI method. It would not only predict a drug, but also explain why it may work, which patient subgroup may benefit, and how confident the prediction is. This can help make AI-based cancer drug repurposing more useful for real clinical practice.

VII. CONCLUSION

AI has made cancer drug repurposing faster and smarter. Recent methods include QSAR, SVM, deep learning, network medicine, single-cell analysis, graph neural networks, and hybrid computational-experimental pipelines. However, the main weakness is that many predictions are not yet strongly validated in patient-like models. Therefore, future research should focus on explainable, patient-specific, and experimentally validated AI pipelines. This would help move cancer drug repurposing from computer prediction toward real patient benefit.

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